

Prestress Force.

$$P_g = 14 \times 250 \times 0.75 = 2625 \text{ kN} //$$

$$P_i = 0.88 \times 2625 = 2310 \text{ kN} //$$

$$P_e = 2310 \times 0.94 = 2171.4 \text{ kN} //$$

(a)

Check if the section @ mid-span is cracked [2 points]

For $\psi_s = 1.0$

$$W_s = G + \psi_s Q = [(25 \times 0.363) + 2.0] + 1.0 \times 20$$

$$W_s = 31.075 \text{ kN/m.}$$

$$M_s = \frac{W_s l^2}{8} = \frac{31.075 \times 20^2}{8} = 1554 \text{ kNm} //$$

For $\psi_s = 0.7$

$$W_s = G + \psi_s Q = [(25 \times 0.363) + 2.0] + 0.7 \times 20$$

$$W_s = 25.075 \text{ kN/m.}$$

$$M_s = \frac{W_s l^2}{8} = \frac{25.075 \times 20^2}{8} = 1253.75 \text{ kNm} //$$

$$f_{cf}' = 0.6 \times \sqrt{32} = 3.39$$

$$\rho_w = \left(\frac{A_{st} + A_{pt}}{b_w \times d_{ef}} \right) = \frac{1250 + (14 \times 143)^{2002}}{150 \times \left(\frac{d_p + d_{st}}{2} \right)}$$

$$\rho_w = \frac{3252}{150 \times \left(\frac{1000 + 1150}{2} \right)} = 0.02 //$$

0.5

$$\sigma_{cs} = \frac{(2.5 \rho_w - 0.8 \rho_w) E_s \times E_{cs}^*}{(1 + 50 \rho_w)}$$

$$= \left(\frac{2.5 \times 0.02 - 0}{1 + 50 \times 0.02} \right) \times 200 \times 10^3 \times 545 \times 10^{-6}$$

$$\sigma_{cs} = 2.7 //$$

0.5

Cracking Moment (cl. 8.5.3.1)

$$Z = \frac{62 \times 10^9}{650} = 95.38 \times 10^6 \text{ mm}^3.$$

0.5

$$M_{cr} = Z \left(f_{cf}' - \sigma_{cs} + \frac{P_e}{A_g} \right) + P_e \times e.$$

$$= 95.38 \times 10^6 \left(3.39 - 2.7 + \frac{2171.4 \times 10^3}{363 \times 10^3} \right)$$

$$+ 2171.4 \times 10^3 \times (650 - 220)$$

$$M_{cr} = 1570 \text{ kNm} //$$

0.5

(b) Ultimate Bending Moment Capacity (ϕM_u)

(2)

Determine d_n (2 points)

Check whether stress block extends to web
(assume steel has yielded)

$$T_s = A_s f_y = 1250 \times 500 = 625 \times 10^3 \text{ N}$$

$$T_p = A_p f_{py} = 2002 \times 1790 = 3584 \times 10^3 \text{ N}$$

(0.5)

$$\begin{aligned} \text{Compression force in flange } (C_f) &= 0.8 \times f_c' \times (b \times t) \\ &= 0.8 \times 32 \times (240 \times 600) \\ &= 3686 \times 10^3 \text{ N} \end{aligned}$$

(0.25)

$$(T_s + T_p) = 4209 \times 10^3 \text{ N} > (C_f) = 3686 \times 10^3 \text{ N}$$

$\Rightarrow \therefore$ stress block must extend into web.

(0.25)

Calculate Compression Force in flange and Web.

$$\begin{aligned} C_f &= \alpha \times f_c' \times t \times (b - b_w) \\ &= 0.8 \times 32 \times 240 \times (600 - 150) \end{aligned}$$

$$C_f = 2765 \text{ kN}$$

(0.25)

$$C_w = \alpha f_c' \gamma d_n b_w$$

$$= 0.8 \times 32 \times 0.89 d_n \times 150$$

$$\gamma = 0.97 - 0.0025 f_c$$

$$\gamma = 0.97 - 0.0025 \times 32$$

$$\gamma = 0.89 //$$

$$C_w = 3418 d_n //$$

0.25

Considering the equilibrium.

$$C_s + C_w = T_s + T_p$$

$$2765 + 3418 d_n = 4209 \times 10^3$$

$$d_n = 422 \text{ mm} //$$

0.5

Check assumption of steel has yielded.

$$\epsilon_{st} = \frac{\epsilon_{cp} (d_{st} - d_n)}{d_n} = \frac{0.003 \times (1150 - 422)}{422}$$

$$\epsilon_{st} = 0.0051 > 0.0025$$

$\therefore$ yielded.

Calculate Strain in Prestressing Steel.

0.25

$$\epsilon_p = \epsilon_{ce} + \epsilon_{pe} + \epsilon_{cp}$$

$$\sigma_{ce} = \frac{P_c}{A_g} + \frac{P_c e^2}{I_g} = \frac{2171.4}{363 \times 10^3} + \frac{2171 \times (650 - 220)^2}{62 \times 10^9}$$

$$\sigma_{ce} = 12.46 \text{ MPa}$$

0.25

$$\epsilon_{ce} = \frac{\sigma_{ce}}{E_c} = \frac{12.46}{30100} = 0.000414 //$$

(0.25)

$\epsilon_{pe} \Rightarrow$ due to strain in prestress steel.

$$\epsilon_{pe} = \frac{\sigma_{pe}}{E_p} = \frac{(2171 \times 10^3) \times 1/2002}{190,000} = 0.0057 //$$

(0.25)

$\epsilon_{cp} =$ Strain in concrete @ level of ~~concrete~~ ^{prestress} due to applied M.

$$\epsilon_{cp} = \frac{\epsilon_{cu} (d_p - d_n)}{d_n} = \frac{0.003 \times (1000 - 422)}{422}$$

$$\epsilon_{cp} = 0.0041$$

(0.25)

Total Strain at the level of Prestress (ϵ_p)

$$\epsilon_p = \epsilon_{ce} + \epsilon_{pe} + \epsilon_{cp}$$

$$= 0.000414 + 0.0057 + 0.0041$$

$$\therefore \epsilon_p = 0.01 > 0.009 \quad \text{OK.}$$

(0.25)

Calculate Moment Capacity (2 points)

$$M_u = T_s d_s + T_p d_p + C_f d_f - C_w d_w$$

$$= (625 \times 10^3 \times 1150) + (3584 \times 10^3 \times 1000) - (2765 \times 10^3 \times 120) - (3418 \times 422) \times \left(\frac{422 \times 0.89}{2} \right)$$

$d_w = \left(\frac{2d_w}{2} \right)$
↑

$$M_u = 3700 \text{ kNm}$$

0.25

Ultimate Bending Moment Capacity $(\phi M_u) = 3700 \times 0.85$

$$(\phi M_u) = 3145 \text{ kNm}$$

0.25

(4)

(3)

Design Shear Force @ $x = 1.0\text{m}$ Design Load

$$W = 1.20 G_k + 1.5 Q_k$$

$$W = 1.2(25 \times 0.363 + 2.0) + 1.5 \times 20$$

$$W = 43.30 \text{ kN/m}$$

Max^m Moment & Shear Force

$$M = \frac{Wl^2}{8} = \frac{43.30 \times 20^2}{8} = 2165 \text{ kNm}$$

$$V = \frac{Wl}{2} = \frac{43.30 \times 20}{2} = 433 \text{ kN}$$

Shear Force & BM @ $x = 1.0\text{m}$ Shear Force

$$\frac{433}{10} = \frac{V_x}{(10-x)}$$

$$V_x = 433 - 43.3x //$$

$$\text{When } x = 1.0; \quad V_x = 389.7 \text{ kN} //$$

0.25

Bending Moment

$$M_x = V_x \times x + \frac{Wx^2}{2}$$

$$= (433 - 43.3x)x + 43.3 \frac{x^2}{2} \quad (0.25)$$

$$M_x = 433x + 21.65x^2 \Rightarrow M(1) = 411 \text{ kNm}$$

Cable Eccentricity

$$y(x) = 4h \left[\frac{x}{L} - \left(\frac{x}{L} \right)^2 \right]$$

$$h = 660 - 220$$

$$h = 440$$

$$e(x) = 4 \times 0.44 \left[\frac{x}{20} - \left(\frac{x}{20} \right)^2 \right]$$

$$e(x) = 1.76 \left[\frac{x}{20} - \frac{x^2}{400} \right] //$$

Slope of the Cable

$$\frac{d}{dx} [y(x)] = \frac{4h}{L} \left[1 - 2\frac{x}{L} \right]$$

$$\theta = \frac{4 \times 0.44}{20} \left[1 - \frac{2x}{20} \right]$$

$$\theta = 0.088 \left[1 - 0.1x \right] //$$

Effective Width of Web

$$b_v = (b_w - 0.5 \sum d_a)$$

$$= 150 - (0.5 \times 112) = 94 \text{ mm.} //$$

0.25

Effective Depth of Section

$$d = \left[\frac{A_{st} d_{st} + A_p d_p}{A_{st} + A_p} \right]$$

$$d_p = e(1) + 0.55$$

$$e(1) = 1.76 \left[\frac{1}{20} - \frac{1^2}{400} \right] = 0.0836 \text{ m} //$$

So;

$$\Rightarrow d_p = 0.0836 + 0.55 = 0.634 \text{ m} //$$

Thus;

$$d = \frac{(1250 \times 1150) + (2002) \times 634}{1250 + 2002} = 832 \text{ mm} //$$

Internal Lever Arm

$$d_v = \max^m (0.72D ; 0.9d)$$

$$= \max^m (0.72 \times 1200 ; 0.9 \times 832)$$

$$d_v = \max^m (864 ; 748) = 864 \text{ mm} //$$

Calculate Concrete Contribution to Shear Capacity
(Cl. 8.2.4.1) - [2 points.]

$$\alpha = 0.088 [1 - 0.1\alpha]$$

$$\alpha(1) = 0.0792 //$$

$$P_v = P\alpha = 2171.4 \times 0.0792$$

$$P_v = 172 \text{ kN} //$$

$$\epsilon_x = \frac{\left(\frac{M^*}{d_v}\right) + (V^*) - \gamma_p P_v + 0.5 N^* - A_{pt} f_{po}}{2 \times [E_s A_{st} + E_p A_{pt}]}$$

$$= \frac{\frac{41 \times 10^6}{864} + 389.7 \times 10^3 - 0.9 \times 172 \times 10^3 - 2002 \times 0.5 \times 1790}{2 \times [200 \times 10^3 \times 1250 + 190,000 \times 2002]}$$

$$\epsilon_x < 0;$$

So,

$$\epsilon_x = \frac{\left(\frac{M^*}{d_v}\right) + (V^*) - \gamma_p P_v + 0.5 N^* - A_{pt} f_{po}}{2 \times [E_s A_{st} + E_p A_{pt} + E_c A_{ct}]}$$

$$= \frac{-1081195.55}{2 \times [200 \times 10^3 \times 1250 + 190 \times 10^3 \times 2002 + 30100 \times (300 \times 400 + 150 \times 350)]}$$

$$\epsilon_x = -0.0000928 \quad (0.25)$$

$$\text{Thus, } k_v = \frac{0.4}{1 + 1500 \times 9.28 \times 10^{-5}} = 0.351 \quad (0.25)$$

$$Q_v = 29 + 7000 \times \epsilon_x = 29 + 7000 \times 9.28 \times 10^{-5} = 29.65 \quad (0.25)$$

$$V_{uc} = k_v b_v d_v \sqrt{f_c'} = 0.351 \times 94 \times 882 \times \sqrt{32} = 155.3 \text{ kN}$$

$$\phi V_{uc} = 116.5 \text{ kN} \quad (0.25)$$

Design Shear Force - R/f (1 Mark)

⑥

Minimum shear required to take by stirrup

$$V_{us} = \frac{1}{\phi} \left[V^* - (\phi V_{uc} + \gamma_p P_v) \right]$$
$$= \frac{1}{0.75} \left[389.7 - (0.75 \times 155.3 + 0.9 \times 172) \right]$$

$$V_{us} = 158 \text{ kN}$$

0.25

Area of shear r/f per spacing

$$\frac{A_{sv}}{s} = \frac{V_{us}}{f_{sy,s} d_v \cot \theta_v}$$

$$\left(\frac{A_{sv}}{s} \right) = \frac{158 \times 10^3}{250 \times 832 \times \cot (29.65^\circ)}$$

$$\left(\frac{A_{sv}}{s} \right) = 0.432$$

0.25

Check min^m rweb r/f

$$\left(\frac{A_{sv, \min}}{s} \right) = \frac{0.08 \times \sqrt{f_c'} b_v}{f_{sy,s}} = \frac{0.08 \times \sqrt{32} \times 94}{250}$$

$$\left(\frac{A_{sv, \min}}{s} \right) = 0.17 //$$

$$\left(\frac{A_{sv}}{s}\right) > \left(\frac{A_{sv,min}}{s}\right)$$

Hence:- $\left(\frac{A_{sv}}{s}\right) = 0.432$

Take $A_{sv} = 80 \times 2 = 160 \text{ mm}^2$

$$s = \frac{160}{0.432} = 370 \text{ mm} //$$

0.25

However;

$$s = \min^m \left\{ s, \min^m \text{ spacings} = 300 \text{ mm} \right\}$$

$$s = 300 \text{ mm} //$$

$\therefore$ adopt R10 @ 300 c/c.

0.25